

1. Let $x(n)$, $0 \leq n \leq N-1$ has DFT $X(k)$; $0 \leq k \leq N-1$

(a) If $x(n)$ is symmetric, i.e., $x(n) = x(N-1-n)$

Show that $X\left(\frac{N}{2}\right) = 0$ for 'N' even.

(b) If $x(n)$ is anti symmetric, i.e., $x(n) = -x(N-1-n)$

Show that $X(0) = 0$

(c) If $x(n) = -x(n+M)$; $N = 2M$.

Show that $X(2l) = 0$ for $l = 0, \dots, M-1$

2. Consider 12 length seq. defined for $0 \leq n \leq 11$

$$x(n) = \{3 \ -1 \ 2 \ 4 \ -3 \ -2 \ 0 \ 1 \ -4 \ 6 \ 2 \ 5\}$$

With 12-pt DFT given by $X(k)$; $0 \leq k \leq 11$. Evaluate without computing the DFT:

(a) $X(0)$ (b) $\sum_{k=0}^{11} e^{-j(4\pi k/6)} X(k)$ (c) $\sum_{k=0}^{11} |X(k)|^2$

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